

Section 5.2 – 5.3 Exercises

Directions: For exercises 1 - 4, use the given sign of the sine and cosine functions to find the quadrant in which the terminal point determined by t lies.

1. $\sin(t) < 0$ and $\cos(t) < 0$
2. $\sin(t) > 0$ and $\cos(t) > 0$
3. $\sin(t) > 0$ and $\cos(t) < 0$
4. $\sin(t) < 0$ and $\cos(t) > 0$

Directions: For exercises 5 - 17, find the exact value of each trigonometric function.

5. $\sin \frac{\pi}{2}$
6. $\sin \frac{\pi}{3}$
7. $\cos \frac{\pi}{2}$
8. $\cos \frac{\pi}{3}$
9. $\sin \frac{\pi}{4}$
10. $\cos \frac{\pi}{4}$
11. $\sin \frac{\pi}{6}$
12. $\sin \pi$
13. $\sin \frac{3\pi}{2}$
14. $\cos \pi$
15. $\cos 0$
16. $\cos \frac{\pi}{6}$
17. $\sin 0$

Directions: For exercises 18 - 28, state the reference angle for the given angle.

18. 240°
19. -170°
20. 100°
21. -315°
22. 135°

23. $\frac{5\pi}{4}$

24. $\frac{2\pi}{3}$

25. $\frac{5\pi}{6}$

26. $\frac{-11\pi}{3}$

27. $\frac{-7\pi}{4}$

28. $\frac{-\pi}{8}$

Directions: For exercises 29 - 42, find the reference angle, the quadrant of the terminal side, and the sine and cosine of each angle.

29. 225°

30. 300°

31. 135°

32. 210°

33. 120°

34. 150°

35. $\frac{5\pi}{4}$

36. $\frac{7\pi}{6}$

37. $\frac{5\pi}{3}$

38. $\frac{3\pi}{4}$

39. $\frac{4\pi}{3}$

40. $\frac{2\pi}{3}$

41. $\frac{5\pi}{6}$

42. $\frac{7\pi}{4}$

Directions: For exercises 43 - 46, find the requested value using the given information.

43. If $\cos(t) = \frac{1}{7}$ and t is in the 4th quadrant, find $\sin(t)$.

44. If $\cos(t) = \frac{2}{9}$ and t is in the 1st quadrant, find $\sin(t)$.

45. If $\sin(t) = \frac{3}{8}$ and t is in the 2nd quadrant, find $\cos(t)$.

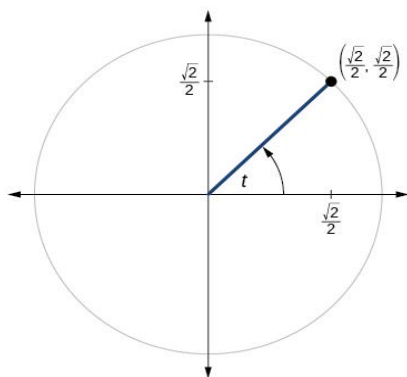
46. If $\sin(t) = -\frac{1}{4}$ and t is in the 3rd quadrant, find $\cos(t)$.

47. State the domain of the sine and cosine functions in interval notation.

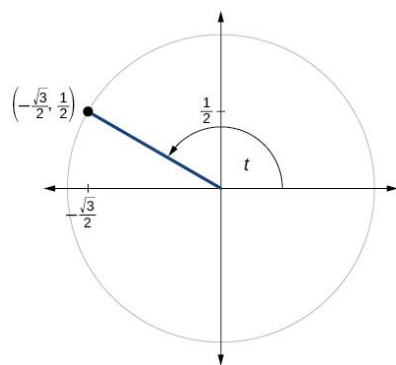
48. State the range of the sine and cosine functions in interval notation.

Directions: For exercises 49 - 64, use the given point on the unit circle to find the value of the sine and cosine of t .

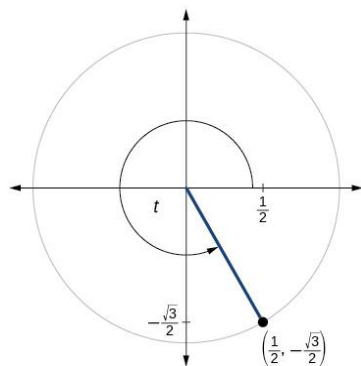
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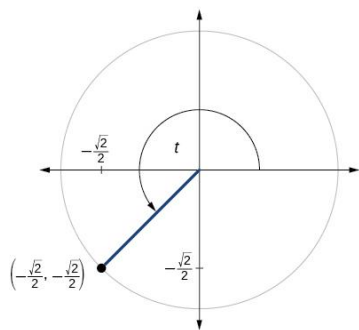
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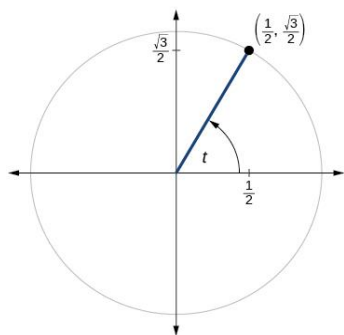
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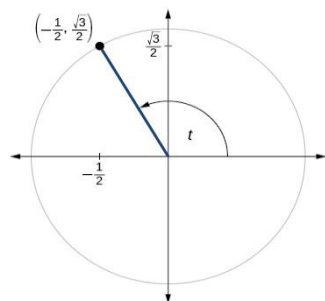
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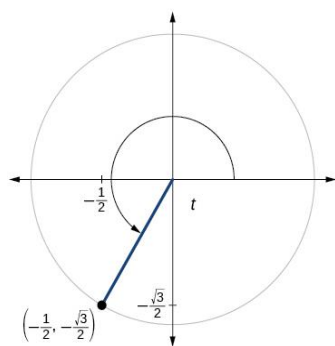
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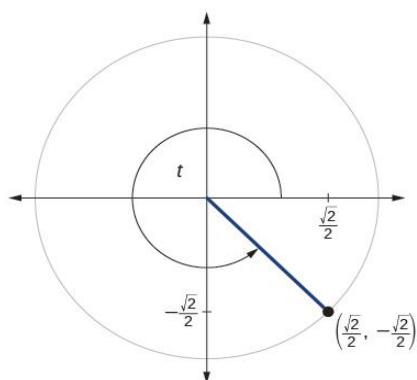
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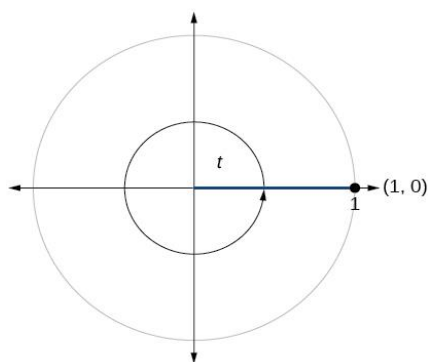
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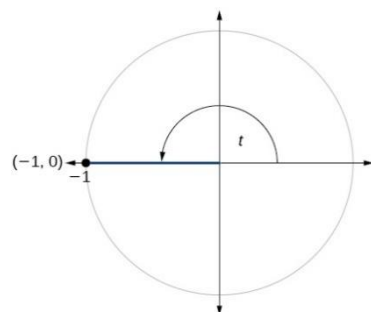
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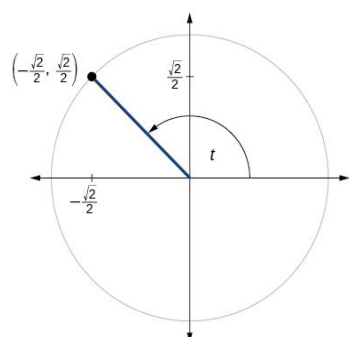
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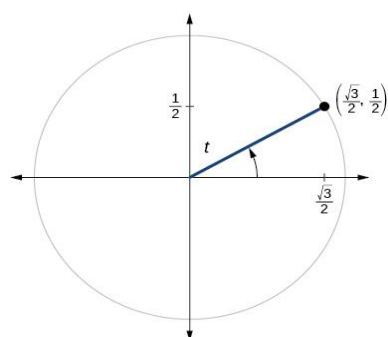
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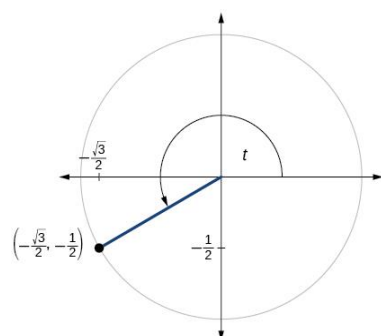
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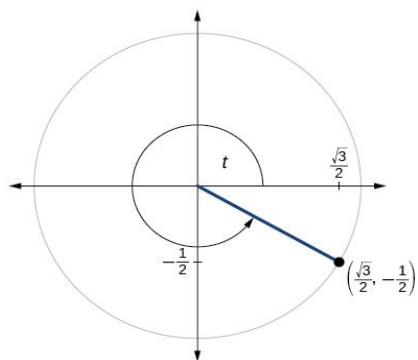
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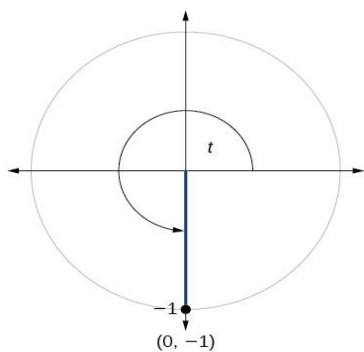
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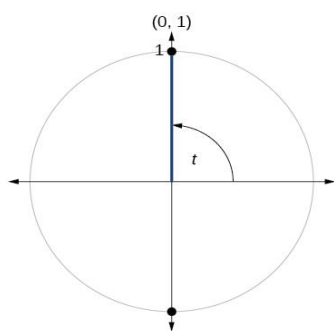
62.



63.



64.



Directions: For exercises 65 - 96, find the exact value of each expression.

65. $\tan \frac{\pi}{6}$

66. $\sec \frac{\pi}{6}$

67. $\csc \frac{\pi}{6}$

68. $\cot \frac{\pi}{6}$

69. $\tan \frac{\pi}{4}$

70. $\sec \frac{\pi}{4}$

71. $\csc \frac{\pi}{4}$

72. $\cot \frac{\pi}{4}$

73. $\tan \frac{\pi}{3}$

74. $\sec \frac{\pi}{3}$

75. $\csc \frac{\pi}{3}$

76. $\cot \frac{\pi}{3}$

77. $\tan \frac{5\pi}{6}$

78. $\sec \frac{7\pi}{6}$

79. $\csc \frac{11\pi}{6}$

80. $\cot \frac{13\pi}{6}$

81. $\tan \frac{7\pi}{4}$

82. $\sec \frac{3\pi}{4}$

83. $\csc \frac{5\pi}{4}$

84. $\cot \frac{11\pi}{4}$

85. $\tan \frac{8\pi}{3}$

86. $\sec \frac{4\pi}{3}$

87. $\csc \frac{2\pi}{3}$

88. $\cot \frac{5\pi}{3}$

89. $\tan 225^\circ$

90. $\sec 300^\circ$

91. $\csc 150^\circ$

92. $\cot 240^\circ$

93. $\tan 330^\circ$

94. $\sec 120^\circ$

95. $\csc 210^\circ$

96. $\cot 315^\circ$

97. If $\sin t = \frac{3}{4}$, and t is in quadrant 2, find $\cos t$, $\sec t$, $\csc t$, $\tan t$, $\cot t$.

98. If $\cos t = -\frac{1}{3}$, and t is in quadrant 3, find $\sin t$, $\sec t$, $\csc t$, $\tan t$, $\cot t$.

99. If $\tan t = \frac{12}{5}$, and $0 \leq t < \frac{\pi}{2}$, find $\sin t, \cos t, \sec t, \csc t, \cot t$.

100. If $\sin t = \frac{\sqrt{3}}{2}$ and $\cos t = \frac{1}{2}$, find $\sec t, \csc t, \tan t, \cot t$.

101. If $\sin t = \frac{\sqrt{2}}{2}$, what is the $\sin(-t)$?

102. If $\cos t = \frac{1}{2}$, what is the $\cos(-t)$?

103. If $\sec t = 3.1$, what is the $\sec(-t)$?

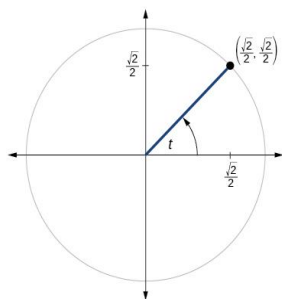
104. If $\csc t = 0.34$, what is the $\csc(-t)$?

105. If $\tan t = -1.4$, what is the $\tan(-t)$?

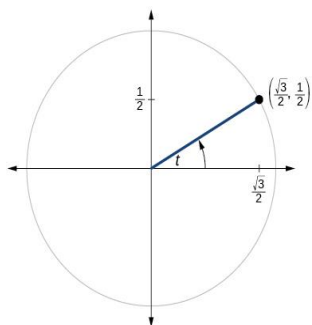
106. If $\cot t = 9.23$, what is the $\cot(-t)$?

Directions: For exercises 107 - 109, use the angle in the unit circle to find the exact value of the each of the six trigonometric functions.

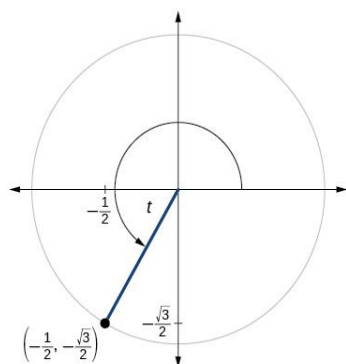
107.



108.



109.



Directions: For exercises 110 - 119, use a calculator to evaluate each expression. Round all answers to three decimal places.

110. $\csc \frac{5\pi}{9}$

111. $\cot \frac{4\pi}{7}$

112. $\sec \frac{\pi}{10}$

113. $\tan \frac{5\pi}{8}$

114. $\sec \frac{3\pi}{4}$

115. $\csc \frac{\pi}{4}$

116. $\tan 98^\circ$

117. $\cot 33^\circ$

118. $\cot 140^\circ$

119. $\sec 310^\circ$